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24CS206 VFSTR

CSE - II Year II Semester

COURSE SYLLABUS

Course Code	Course Title	L	T	P	SL	C
25CY206	Statistical Foundation for Computing and Data Science	3	0	2	3	4
NEW CODE	Web Technologies	0	2	2	0	2

Formatted to follow the supplied university syllabus/PDF structure.

25CY206 - STATISTICAL FOUNDATION FOR COMPUTING AND DATA SCIENCE

Hours Per Week :

L	T	P	SL	C
3	0	2	3	4

PREREQUISITE KNOWLEDGE: Basic knowledge in Mathematics and Statistics.

COURSE DESCRIPTION: This course introduces students to probability theory, statistical inference, and computational statistical techniques essential for engineering, Artificial Intelligence, Machine Learning, Data Science, and scientific computing applications. It provides a rigorous foundation in random variables, probability distributions, sampling theory, estimation, hypothesis testing, regression analysis, and multivariate statistical methods. The course emphasizes statistical reasoning under uncertainty and data-driven decision-making. It also integrates computational tools and AI-based platforms for visualization, simulation, and analysis of statistical models. The course enables students to develop analytical thinking, mathematical modeling skills, and computational proficiency required for modern data-centric engineering systems and intelligent computing applications.

COURSE OBJECTIVES:

Understand fundamental probability theory and random variables.

Apply probability distributions in computing and data science problems.

Develop statistical inference and estimation techniques.

Perform hypothesis testing for engineering decision-making.

Analyze relationships using regression models.

Introduce PCA and time series forecasting for AI systems.

Strengthen computational and AI-assisted statistical skills.

MODULE-1

UNIT-1

Introduction to probability and its applications in computing, probability axioms, conditional probability and independence, Bayes' theorem and its applications. Prior probability and posterior probability; Bayesian inference; Maximum A Posteriori (MAP) estimation.

UNIT-2

Random variables: distribution function, discrete random variable, probability mass function, continuous random variable, probability density function, expectation, variance and their properties. Discrete probability distributions: Uniform, Binomial, Poisson. Continuous probability distributions: Uniform, Normal, Exponential, Weibull, Cauchy.

PRACTICES:

- Explore axioms of probability by formulating real-life event spaces and verifying probability rules.
- Visualize conditional probability using Venn diagrams and real-world scenarios.
- Analyze Bayes' theorem in decision-making problems such as diagnosis and classification.
- Compare discrete and continuous random variables using PMF and PDF representations.
- Investigate Binomial and Poisson distributions using real-world models.
- Compute expectation, variance, and higher moments for datasets.
- Construct probability trees for multi-stage experiments.
- Simulate random variables from different probability distributions.
- Analyze Normal and Exponential distributions using real-world datasets.
- Estimate distribution parameters and visualize probability distributions.

MODULE-2

UNIT-1

Population and sample, sampling distributions, point estimation and properties of estimators, introduction to Maximum Likelihood Estimation (MLE), and confidence intervals for population parameters. Hypothesis testing: null and alternative hypotheses, level of significance, p-value, critical value, Type I and Type II errors, single-sample and two-sample tests for means and variance, and chi-square test for independence of attributes. A/B Testing and its applications in data-driven decision-making and AI-enabled systems.

UNIT-2

Correlation analysis: scatter plots, Pearson's correlation coefficient, and interpretation. Simple linear regression model and assumptions, regression coefficients, fitting and interpretation of the regression line, coefficient of determination (R^2), prediction, and residual analysis. Introduction to data transformations (log and square root) and logistic regression for binary classification.

UNIT-3

Introduction to Multivariate Analysis (MVA), multivariate data, covariance and correlation matrices, Principal Component Analysis (PCA), dimensionality reduction, interpretation and applications of principal components. Introduction to Time Series Analysis, time series data, components of a time series, trend estimation using least squares, moving averages, and simple exponential smoothing for forecasting.

PRACTICES:

- Analyze sampling distributions and estimate population parameters.
- Explore Maximum Likelihood Estimation by interpreting likelihood functions.
- Construct and interpret confidence intervals for engineering datasets.
- Perform hypothesis testing using Z-test, t-test, and chi-square test.
- Compare parametric tests based on assumptions and applicability.
- Visualize simple linear regression models and residual distributions.
- Analyze logistic regression for binary classification problems.
- Apply PCA for dimensionality reduction and interpret principal components.
- Investigate time-series data patterns including trend and seasonality.
- Validate statistical inference results and compare theoretical versus computed conclusions.

COURSE OUTCOMES:

CO	Course Outcomes	Bloom's Level	POs
CO1	Understand probability, random variables, and distributions and apply them in computing applications.	Apply	1, 2, 12
CO2	Analyze probability distributions, their properties, and significance in computing and machine learning.	Analyze	1, 2, 4

SKILLS:

- ✓ *Statistical reasoning and probabilistic modeling.*
- ✓ *Analytical thinking for engineering problem solving.*
- ✓ *Mathematical modeling of real-world systems.*
- ✓ *Data-driven decision-making ability.*
- ✓ *Machine learning and AI-assisted statistical analysis.*
- ✓ *Regression and predictive modeling techniques.*
- ✓ *Dimensionality reduction and feature engineering.*
- ✓ *Time series forecasting and trend analysis.*
- ✓ *Python-based scientific computing skills.*
- ✓ *Data visualization and interpretation skills.*
- ✓ *NEW CODE - WEB TECHNOLOGIES*
- ✓ *Hours Per Week*
- ✓ *PREREQUISITE: Object-Oriented Programming (Java or Python)*

CO	Course Outcomes	Bloom's Level	POs
CO3	Perform statistical inference using estimation and hypothesis testing techniques and interpret results.	Evaluate	2, 3, 5
CO4	Apply linear and logistic regression techniques and interpret results.	Apply	2, 4, 6
CO5	Evaluate PCA and Time Series Analysis for data-driven decision-making.	Evaluate	3, 5, 11

TEXT BOOKS:

- Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press, 2019.
- Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage Learning, 2015.

REFERENCE BOOKS:

- Morris H. DeGroot and Mark J. Schervish, Probability and Statistics, 4th Edition, Pearson, 2011.
- Casella, George and Roger L. Berger, Statistical Inference, 2nd Edition, Cengage Learning, 2002.
- Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.

Hours Per Week :

L	T	P	SL	C
0	2	2	0	2

- COURSE DESCRIPTION: This course bridges the gap between traditional full-stack web development and modern artificial intelligence. It offers comprehensive training in building dynamic web applications using HTML5, CSS3, modern JavaScript (ES6+), and Node.js, while seamlessly integrating AI capabilities. Students will learn to leverage AI coding assistants, implement prompt engineering, integrate Large Language Model (LLM) APIs, and utilize vector databases. The ultimate objective is to empower students to design, develop, and deploy intelligent, agentic web applications that can understand, process, and generate dynamic content.

MODULE-1: MODERN WEB FOUNDATIONS & CLIENT-SIDE INTELLIGENCE

- Flow Logic: We start by building static pages, add logic with JavaScript, introduce modern frontend frameworks, and then inject AI directly into the user's browser.

UNIT-1 **0L+6T+6P+0SL=12 Hours**

CORE WEB TECHNOLOGIES & AI-ASSISTED DEVELOPMENT

- Modern HTML5 & CSS3: Semantic HTML, Canvas, Audio and Video Integration, CSS3 Grid, Flexbox, Responsive Design, CSS Variables. JavaScript (ES6+): DOM Manipulation, Arrow Functions, Promises, Async/Await, Closures, Fetch API, Event Handling, Regular Expression Validation. Code Debugging, Boilerplate Code Generation.

UNIT-2 **0L+6T+6P+0SL=12 Hours**

CLIENT-SIDE FRAMEWORKS & BROWSER AI

- Frontend Frameworks: Component Architecture, State Management, Props and Data Binding, Routing (Angular/React). Intelligent UI/UX: Conversational Interfaces, Chat UI, Adaptive Dynamic Forms. Browser-Based AI: Web Speech API, Speech-to-Text, Text-to-Speech, Browser-based Pre-trained Models, TensorFlow.js, Image Classification, Client-side Sentiment Analysis.

PRACTICES:

- Design a responsive, multi-page personal portfolio using HTML5 and CSS3 with AI-assisted scaffolding.
- Develop an interactive catalog page using JavaScript with client-side filtering and sorting.
- Design a comprehensive student registration form with JavaScript validation and lightweight client-side AI.
- Create a voice-controlled web page using the Web Speech API.

• Develop a web application that captures a webcam image and uses a pre-trained TensorFlow.js model to identify the primary object.

MODULE-2: SERVER-SIDE DATA & INTELLIGENT WEB AGENTS

• Flow Logic: We build a secure backend server first, then connect that server to LLMs and Vector DBs to give it intelligence, and finally upgrade the application to act as an autonomous agent before deploying.

UNIT-1 **0L+6T+6P+0SL=12 Hours**

SECURE BACKEND DEVELOPMENT & DATABASES

• Node.js & Express: Node.js Server Setup, Express Routing, Middleware, RESTful APIs, CORS. Authentication & Sessions: JSON Web Tokens (JWT), Secure Login/Signup Flows, Protected Routes, Session Management. Database Integration: NoSQL Fundamentals, MongoDB, CRUD Operations, Mongoose ODM.

UNIT-2 **0L+6T+6P+0SL=12 Hours**

LLM INTEGRATION & CONTEXTUAL MEMORY

• API Integration: OpenAI API, Google Gemini API, Secure API Key Management, Environment Variables. Prompt Engineering: System Prompts, Few-shot Prompting, Context Window Management, AI Hallucination Mitigation. Vector Databases & RAG: Data Embeddings, Vector Database Setup (Pinecone/Weaviate/MongoDB Atlas Vector Search), Semantic Data Storage.

UNIT-3 **0L+6T+6P+0SL=12 Hours**

BUILDING AGENTIC TOOLS & DEPLOYMENT

• Agentic Web Workflows: Autonomous AI Tasks, Tool Calling, Automated Web Scraping, Content Summarization Pipelines. Deployment & Security: Application Hosting (Vercel/AWS EC2), Secure Environment Configurations, Prompt Injection Security, Endpoint Rate Limiting.

PRACTICES:

• Design a RESTful API using Node.js and Express connected to MongoDB for complete CRUD operations.

• Implement secure login and signup using JSON Web Tokens (JWT) and protected routes.

• Develop a frontend chatbot connected to a Node.js backend and an LLM API.

• Design a semantic search engine using embeddings and a vector database.

• Develop an intelligent web agent capstone that accepts a document or URL, extracts text, summarizes it, translates it, and renders results on a secure dashboard.

COURSE OUTCOMES:

CO	Course Outcomes	Bloom's Level	Module / Unit	POs
CO1	Understand foundational modern web technologies including semantic HTML5, CSS3, JavaScript (ES6+) and AI-assisted development tools.	Understand	M-I U-I	PO1, PO2, PO5
CO2	Describe component architecture of modern frontend frameworks and browser-based AI APIs such as Web Speech and TensorFlow.js.	Understand	M-I U-II	PO1, PO2
CO3	Apply Node.js, Express and NoSQL databases to build and secure server-side RESTful APIs with JWT authentication.	Apply	M-II U-I	PO1, PO2, PO3, PO5
CO4	Analyze prompt engineering strategies and Retrieval-Augmented Generation (RAG) architectures for integrating LLMs and vector databases.	Analyze	M-II U-II	PO1, PO2, PO3, PO4
CO5	Design and deploy comprehensive multi-tier intelligent web applications that execute autonomous agentic workflows while addressing endpoint security.	Create	M-II U-III	PO2, PO3, PO5, PO8

• TEXTBOOKS:

• Brad Dayley, Node.js, MongoDB and Angular Web Development, 2nd Edition, Addison-Wesley Professional, 2017.

• Jon Duckett, HTML and CSS: Design and Build Websites & JavaScript and JQuery: Interactive Front-End Web Development, Wiley.

• Gant Laborde, Learning TensorFlow.js: Powerful Machine Learning in JavaScript, O'Reilly Media, 2021.

REFERENCE BOOKS:

• Cay S. Horstmann, Modern JavaScript for the Impatient, Addison-Wesley Professional, 2020.

• Olivier Caelen and Marie-Alice Blete, Developing Apps with GPT-4 and ChatGPT, O'Reilly Media, 2023.

• WEB LINKS & OFFICIAL DOCUMENTATION:

• MDN Web Docs — Standard for HTML, CSS, JavaScript and Web APIs.

• React Official Documentation.

• Google Gemini API Documentation / OpenAI API Documentation.

• LangChain.js Documentation — RAG and agentic workflows.

• Pinecone / MongoDB Atlas Vector Search Documentation.